MPSTME LIBRARY
NMIMS UNIVERSITY

SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2023-2024

Program/s: B.Tech

Year: II Semester:III

Stream/s ;CSBS

Subject: Object Oriented Programming

Time: 3 hrs (10:00am to 1:00pm)

Date: 10 / 02/ 2024

No. of Pages:03

Marks: 100

Re exam (2022-23)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO-1; SO-1; BL-1	a.	Write a C++ program to create a structure named as employee and data members of the structure are emp_id,emp_name, exp,salary. Display the employee details having salary of more than 5 lakhs.	[05]
CO-2; SO-1; BL-6	b.	Illustrate the concept of Array of Objects with an example.	[05]
CO-3; SO-1; BL-4	c.	Difference between Abstraction and Encapsulation (Five points)	[05]
CO-4; SO-2; BL-1	d.	Define file handling use and write about opening and closing of file. help of diagram	[05]
Q2 CO-3; SO-1; BL-1,6	a) b)	Define function template. Write a C++ program to find largest of three elements using template.	[10] [10]

<table>
<tr>
<td></td>
<td></td>
<td>Create a structure teacher with data member teacher name, teacher designation (Associate professor / Assistant Professor) and teacher type (permanent/ Visiting).
Create function readData() to accept the input.
Create function cal_sal() which will calculate the salary of teacher based on designation and type of teacher
<table>
<tr><td>Associate professor</td><td>Permanent</td><td>150000/-</td></tr>
<tr><td>Assistant professor</td><td>Permanent</td><td>100000/-</td></tr>
<tr><td>Associate professor</td><td>Visiting</td><td>No.of hrs*1500</td></tr>
<tr><td>Assistant professor</td><td>Visiting</td><td>No.of hrs * 1000</td></tr>
</table>
Create display () function which will display the data of all the teachers.
Note: Accept the data for 3 teachers</td>
<td></td>
</tr>
<tr>
<td>Q3
CO-2;
SO-1;
BL-4,6</td>
<td>a)

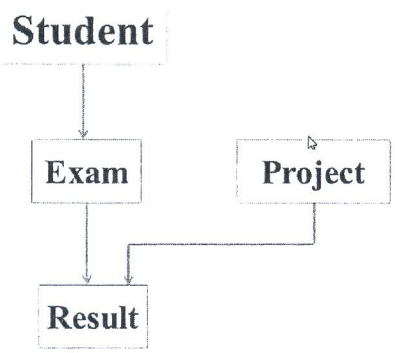
b)</td>
<td>Write a C++ program to calculate the perimeter, area , length of the rectangle using the function overloading concept.
Write a C++ program to generate ‘n’ Armstrong number using inline function concept.

$1634 = 1^4 + 6^4 + 3^4 + 4^4$

Armstrong Number</td>
<td>[10]

[10]</td>
</tr>
<tr>
<td>Q4
CO-3;
SO-1;
BL-6</td>
<td>a)

b)</td>
<td>Write a C++ program to interchange values of two integer numbers using friend function.

Write C++ program for following given information:</td>
<td>[10]

[10]</td>
</tr>
</table>

		1. Class student: name, SAP_ID, branch 2. Class exam: OOP,SE,TCS (marks out of 50) 3. Project: Project_title, project_marks (out of 50) 4. Class result: total_marks scored (class exam + project) Input(), result() and display() are the member functions of the class. Write a program for the given scenario.	
Q5 CO-2,4; SO-1,2; BL-3,6	a) b)	Draw an activity diagram for MPSTME student registration for a seminar Write a C++ program to calculate the area of triangle, rectangle and circle using constructor overloading. The program should be menu driven.	[10] [10]
Q6 CO-2; SO-1; BL-6	a) b)	Class internal_marks having data members are name, roll_no, OOP_marks, SE_Marks and Class external_marks having data members are name, roll_no, OOP_marks, SE_Marks, TCS_Marks. Perfrom addition between internal (out of 50) and external marks (out of 50) of OOP_marks, SE_Marks , TCS_Marks and display the final marks of both the classes using friend class concept. Write four different cases of default arguments. Write Single sum function to add 2 numbers, 3 numbers, and 4 numbers by using the concept of default argument.	[10] [10]
Q7 CO2-; SO-1; BL-6	a) b)	Create a class Time which contains: hours , minutes and second. Write a C++ program using operator overloading for the following: 1. = = to check whether two Times are same or not and display the time. Write a C++ program to calculate the percentage of a student using multi-level inheritance. Accept the marks of three subjects in base class. A class will derived from the above mentioned class which includes a function to find the total marks obtained and another class derived from this class which calculates and displays the percentage of student.	[10] [10]